

Q4 Kinematics with calculus

Name: _____

Class: _____

Date: _____

Time: **372 minutes**

Marks: **309 marks**

Comments:

EXAM PAPERS PRACTICE

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Q1.

A remote-controlled toy car is moving over a horizontal surface. It moves in a straight line through a point A.

The toy is initially at the point with displacement 3 metres from A. Its velocity, $v \text{ m s}^{-1}$, at time t seconds is defined by

$$v = 0.06(2 + t - t^2)$$

- (a) Find an expression for the displacement, r metres, of the toy from A at time t seconds.

(4)

- (b) **In this question use $g = 9.8 \text{ m s}^{-2}$**

At time $t = 2$ seconds, the toy launches a ball which travels directly upwards with initial speed 3.43 m s^{-1}

Find the time taken for the ball to reach its highest point.

(3)

(Total 7 marks)

Q2.

A driver is road-testing two minibuses, A and B, for a taxi company.

The performance of each minibus along a straight track is compared.

A flag is dropped to indicate the start of the test.

Each minibus starts from rest.

The acceleration in m s^{-2} of each minibus is modelled as a function of time, t seconds, after the flag is dropped:

$$\text{The acceleration of A} = 0.138t^2$$

$$\text{The acceleration of B} = 0.024t^3$$

- (a) Find the time taken for A to travel 100 metres.

Give your answer to four significant figures.

(4)

- (b) The company decides to buy the minibus which travels 100 metres in the shortest time.

Determine which minibus should be bought.

(4)

- (c) The models assume that both minibuses start moving immediately when $t = 0$

In light of this, explain why the company may, in reality, make the wrong decision.

(1)

(Total 9 marks)

Q3.

A particle, of mass 400 grams, is initially at rest at the point O.

The particle starts to move in a straight line so that its velocity, $v \text{ m s}^{-1}$, at time t seconds is given by

$$v = 6t^2 - 12t^3 \text{ for } t > 0$$

- (a) Find an expression, in terms of t , for the force acting on the particle.

(3)

- (b) Find the time when the particle next passes through O.

(5)

(Total 8 marks)

Q4.

At time $t = 0$, a parachutist jumps out of an airplane that is travelling horizontally. The velocity, $\mathbf{v} \text{ m s}^{-1}$, of the parachutist at time t seconds is given by:

$$\mathbf{v} = (40e^{-0.2t})\mathbf{i} + 50(e^{-0.2t} - 1)\mathbf{j}$$

The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are horizontal and vertical respectively. Assume that the parachutist is at the origin when $t = 0$. Model the parachutist as a particle.

- (a) Find an expression for the position vector of the parachutist at time t .

(4)

- (b) The parachutist opens her parachute when she has travelled 100 metres horizontally.

Find the vertical displacement of the parachutist from the origin when she opens her parachute.

(4)

- (c) Carefully, explaining the steps that you take, deduce the value of g used in the formulation of this model.

(3)

(Total 11 marks)

Q5.

A particle moves in a horizontal plane. The vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular unit vectors in the horizontal plane. At time t seconds, the velocity of the particle, $\mathbf{v} \text{ m s}^{-1}$, is given by

$$\mathbf{v} = 12 \cos \left(\frac{\pi}{3} t \right) \mathbf{i} - 9t^2 \mathbf{j}$$

- (a) Find an expression for the acceleration of the particle at time t . (2)
- (b) The particle, which has mass 4 kg, moves under the action of a single force, $\mathbf{F}$ newtons.
- (i) Find an expression for the force $\mathbf{F}$ in terms of t . (2)
- (ii) Find the magnitude of $\mathbf{F}$ when $t = 3$. (2)
- (c) When $t = 3$, the particle is at the point with position vector $(4\mathbf{i} - 2\mathbf{j})$ m.
- Find the position vector, $\mathbf{r}$ metres, of the particle at time t . (5)

(Total 11 marks)

Q6.

A particle, of mass 3 kg, moves along a straight line. At time t seconds, the displacement, s metres, of the particle from the origin is given by $s = 8t^3 + 15$

- (a) Find the velocity of the particle at time t . (2)
- (b) Find the magnitude of the resultant force acting on the particle when $t = 2$. (4)

(Total 6 marks)

Q7.

A particle, of mass 10 kg, moves on a smooth horizontal plane. At time t seconds, the acceleration of the particle is given by

$$\{(40t + 3t^2)\mathbf{i} + 20e^{-4t}\mathbf{j}\} \text{ m s}^{-2}$$

where the vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular unit vectors.

- (a) At time $t = 1$, the velocity of the particle is $(6\mathbf{i} - 5e^{-4}\mathbf{j}) \text{ m s}^{-1}$.
- Find the velocity of the particle at time t . (5)
- (b) Calculate the initial speed of the particle. (3)

(Total 8 marks)

Q8.

A car accelerates from rest along a straight horizontal road.

The car's engine produces a constant horizontal force of magnitude 4000 N.

At time t seconds, the speed of the car is $v \text{ m s}^{-1}$, and a resistance force of magnitude $40v$ newtons acts upon the car.

The mass of the car is 1600 kg.

(a) Show that $\frac{dv}{dt} = \frac{100 - v}{40}$.

(2)

(b) Find the velocity of the car at time t .

(6)

(Total 8 marks)

Q9.

A particle, of mass 50 kg, moves on a smooth horizontal plane. A single horizontal force

$$[(300t - 60t^2)\mathbf{i} + 100e^{-2t}\mathbf{j}] \text{ newtons}$$

acts on the particle at time t seconds.

The vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular unit vectors.

(a) Find the acceleration of the particle at time t .

(2)

(b) When $t = 0$, the velocity of the particle is $(7\mathbf{i} - 4\mathbf{j}) \text{ m s}^{-1}$.

Find the velocity of the particle at time t .

(4)

(c) Calculate the speed of the particle when $t = 1$.

(4)

(Total 10 marks)

Q10.

Alice places a toy, of mass 0.4 kg, on a slope. The toy is set in motion with an initial velocity of 1 m s^{-1} down the slope. The resultant force acting on the toy is $(2 - 4v)$ newtons, where $v \text{ m s}^{-1}$ is the toy's velocity at time t seconds after it is set in motion.

(a) Show that $\frac{dv}{dt} = -10(v - 0.5)$.

(2)

- (b) By using $\int \frac{1}{v-0.5} dv = - \int 10 dt$, find v in terms of t . (5)
- (c) Find the time taken for the toy's velocity to reduce to 0.55 m s^{-1} . (3)
- (Total 10 marks)

Q11.

A particle moves in a straight line. At time t seconds, it has velocity $v \text{ m s}^{-1}$, where

$$v = 6t^2 - 2e^{-4t} + 8$$

and $t \geq 0$.

- (a) (i) Find an expression for the acceleration of the particle at time t . (2)
- (ii) Find the acceleration of the particle when $t = 0.5$. (2)
- (b) The particle has mass 4 kg.
Find the magnitude of the force acting on the particle when $t = 0.5$. (1)
- (c) When $t = 0$, the particle is at the origin.
Find an expression for the displacement of the particle from the origin at time t . (4)
- (Total 9 marks)

Q12.

A stone, of mass 5 kg, is projected vertically downwards, in a viscous liquid, with an initial speed of 7 m s^{-1} .

At time t seconds after it is projected, the stone has speed $v \text{ m s}^{-1}$ and it experiences a resistance force of magnitude $9.8v$ newtons.

- (a) When $t \geq 0$, show that

$$\frac{dv}{dt} = -1.96(v - 5)$$

(2)

- (b) Find v in terms of t . (5)
- (Total 7 marks)

Q13.

The velocity of a particle at time t seconds is $\mathbf{v} \text{ m s}^{-1}$, where

$$\mathbf{v} = (4 + 3t^2)\mathbf{i} + (12 - 8t)\mathbf{j}$$

- (a) When $t = 0$, the particle is at the point with position vector $(5\mathbf{i} - 7\mathbf{j}) \text{ m}$.

Find the position vector, $\mathbf{r}$ metres, of the particle at time t .

(4)

- (b) Find the acceleration of the particle at time t .

(2)

- (c) The particle has mass 2 kg.

Find the magnitude of the force acting on the particle when $t = 1$.

(4)

(Total 10 marks)

Q14.

Vicky has mass 65 kg and is skydiving. She steps out of a helicopter and falls vertically. She then waits a short period of time before opening her parachute. The parachute opens at time $t = 0$ when her speed is 19.6 m s^{-1} , and she then experiences an air resistance force of magnitude $260v$ newtons, where $v \text{ m s}^{-1}$ is her speed at time t seconds.

- (a) When $t > 0$:

- (i) show that the resultant downward force acting on Vicky is

$$65(9.8 - 4v) \text{ newtons}$$

(1)

- (ii) show that $\frac{dv}{dt} = -4(v - 2.45)$.

(2)

- (b) By showing that $\int \frac{1}{v - 2.45} dv = - \int 4 dt$, find v in terms of t .

(5)

(Total 8 marks)

Q15.

A particle moves in a horizontal plane under the action of a single force, $\mathbf{F}$ newtons. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively. At time t seconds, the velocity of the particle, $\mathbf{v} \text{ m s}^{-1}$, is given by

$$\mathbf{v} = 4e^{-2t}\mathbf{i} + (6t - 3t^2)\mathbf{j}$$

- (a) Find an expression for the acceleration of the particle at time t . (3)
- (b) The mass of the particle is 5 kg.
- (i) Find an expression for the force $\mathbf{F}$ acting on the particle at time t . (2)
- (ii) Find the magnitude of $\mathbf{F}$ when $t = 0$. (2)
- (c) Find the value of t when $\mathbf{F}$ acts due west. (2)
- (d) When $t = 0$, the particle is at the point with position vector $(6\mathbf{i} + 5\mathbf{j})$ m.
- Find the position vector, $\mathbf{r}$ metres, of the particle at time t . (5)
- (Total 14 marks)

Q16.

A particle moves along a straight line through the origin. At time t , the displacement, s , of the particle from the origin is given by

$$s = 5t^2 + 3 \cos 4t$$

Find the velocity of the particle at time t .

(Total 3 marks)

Q17.

A particle has mass 200 kg and moves on a smooth horizontal plane. A single

horizontal force, $\left(400 \cos \left(\frac{\pi}{2} t \right) \mathbf{i} + 600t^2 \mathbf{j} \right)$ newtons, acts on the particle at time t seconds.

The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Find the acceleration of the particle at time t . (2)
- (b) When $t = 4$, the velocity of the particle is $(-3\mathbf{i} + 56\mathbf{j}) \text{ m s}^{-1}$.
- Find the velocity of the particle at time t . (5)
- (c) Find t when the particle is moving due west. (3)

- (d) Find the speed of the particle when it is moving due west.

(2)

(Total 12 marks)

Q18.

A particle is moving along a straight line. At time t , the velocity of the particle is v .

$$-\frac{\lambda}{v^4}$$

The acceleration of the particle throughout the motion is $-\frac{\lambda}{v^4}$, where λ is a positive constant. The velocity of the particle is u when $t = 0$.

Find v in terms of u , λ and t .

(Total 7 marks)

Q19.

A particle moves along a straight line. At time t , it has velocity v , where

$$v = 4t^3 - 8 \sin 2t + 5$$

When $t = 0$, the particle is at the origin.

Find an expression for the displacement of the particle from the origin at time t .

(Total 4 marks)

Q20.

A particle moves on a horizontal plane, in which the unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

At time t seconds, the position vector of the particle is $\mathbf{r}$ metres, where

$$\mathbf{r} = \left(2e^{\frac{1}{2}t} - 8t + 5 \right) \mathbf{i} + (t^2 - 6t) \mathbf{j}$$

- (a) Find an expression for the velocity of the particle at time t .

(3)

- (b) (i) Find the speed of the particle when $t = 3$.

(2)

- (ii) State the direction in which the particle is travelling when $t = 3$.

(1)

- (c) Find the acceleration of the particle when $t = 3$.

(3)

- (d) The mass of the particle is 7 kg.

Find the magnitude of the resultant force on the particle when $t = 3$.

(3)

(Total 12 marks)

Q21.

A particle moves under the action of a force, $\mathbf{F}$ newtons. At time t seconds, the velocity, $\mathbf{v}$ m s⁻¹, of the particle is given by

$$\mathbf{v} = (t^3 - 15t - 5)\mathbf{i} + (6t - t^2)\mathbf{j}$$

- (a) Find an expression for the acceleration of the particle at time t .

(3)

- (b) The mass of the particle is 4 kg.

- (i) Show that, at time t ,

$$\mathbf{F} = (12t^2 - 60)\mathbf{i} + (24 - 8t)\mathbf{j}$$

(2)

- (ii) Find the magnitude of $\mathbf{F}$ when $t = 2$.

(4)

(Total 9 marks)

Q22.

A particle moves in a straight line and at time t it has velocity v , where

$$v = 3t^2 - 2 \sin 3t + 6$$

- (a) (i) Find an expression for the acceleration of the particle at time t .

(2)

- (ii) When $t = \frac{\pi}{3}$, show that the acceleration of the particle is $2\pi + 6$.

(2)

- (b) When $t = 0$, the particle is at the origin.

Find an expression for the displacement of the particle from the origin at time t .

(4)

(Total 8 marks)

Q23.

A particle moves in a horizontal plane under the action of a single force, $\mathbf{F}$ newtons. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively. At time t seconds, the position vector, $\mathbf{r}$ metres, of the particle is given by

$$\mathbf{r} = (t^3 - 3t^2 + 4)\mathbf{i} + (4t + t^2)\mathbf{j}$$

- (a) Find an expression for the velocity of the particle at time t . (2)
- (b) The mass of the particle is 3 kg.
- (i) Find an expression for $\mathbf{F}$ at time t . (3)
- (ii) Find the magnitude of $\mathbf{F}$ when $t = 3$. (2)
- (c) Find the value of t when $\mathbf{F}$ acts due north. (2)
- (Total 9 marks)

Q24.

A particle moves in a straight line and at time t seconds has velocity $v \text{ m s}^{-1}$, where

$$v = 6t^2 + 4t - 7, \quad t \geq 0$$

- (a) Find an expression for the acceleration of the particle at time t . (2)
- (b) The mass of the particle is 3 kg.
- Find the resultant force on the particle when $t = 4$. (2)
- (Total 4 marks)

Q25.

A particle moves in a straight line and at time t seconds has velocity $v \text{ m s}^{-1}$, where

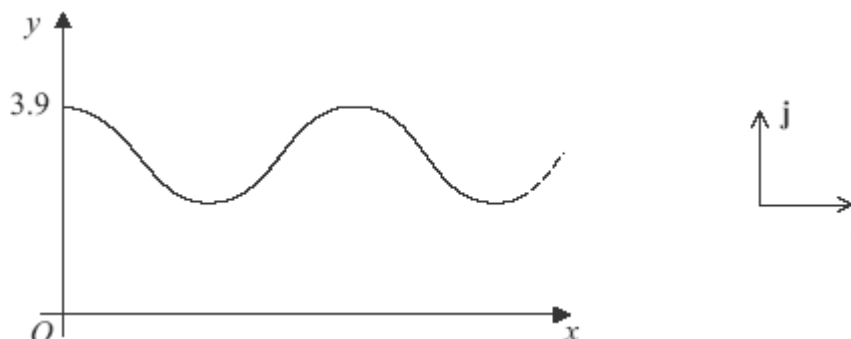
$$v = 6t^2 + 4t - 7, \quad t \geq 0$$

- (a) Find an expression for the acceleration of the particle at time t . (2)
- (b) The mass of the particle is 3 kg.
- Find the resultant force on the particle when $t = 4$. (2)
- (c) When $t = 0$, the displacement of the particle from the origin is 5 metres.
- Find an expression for the displacement of the particle from the origin at time t . (4)

(Total 8 marks)

Q26.

Jane is on a ride in a theme park. Part of the curved path of the ride is shown in the diagram.



Jane's position vector, $\mathbf{r}$ metres, at time t seconds is given by

$$\mathbf{r} = 1.2t \mathbf{i} + (3 + 0.9 \cos t) \mathbf{j}$$

where the perpendicular unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed horizontally and vertically upwards respectively.

- (a) Find an expression for Jane's velocity at time t . (2)
- (b) (i) Find an expression for Jane's speed at time t . (2)
- (ii) Find Jane's maximum speed during the ride. (2)

(Total 6 marks)

Q27.

A particle has mass 800 kg. A single force of $(2400 \mathbf{i} - 4800t \mathbf{j})$ newtons acts on the particle at time t seconds. No other forces act on the particle.

- (a) Find the acceleration of the particle at time t . (2)
- (b) At time $t = 0$, the velocity of the particle is $(6\mathbf{i} + 30\mathbf{j}) \text{ m s}^{-1}$. The velocity of the particle at time t is $\mathbf{v} \text{ m s}^{-1}$.

Show that

$$\mathbf{v} = (6 + 3t)\mathbf{i} + (30 - 3t^2)\mathbf{j}$$

(4)

- (c) Initially, the particle is at the point with position vector $(2\mathbf{i} + 5\mathbf{j}) \text{ m}$.

Find the position vector, $\mathbf{r}$ metres, of the particle at time t .

(5)

(Total 11 marks)

Q28.

A particle moves in a straight line and at time t has velocity v , where

$$v = 2t - 12e^{-t}, \quad t \geq 0$$

- (a) (i) Find an expression for the acceleration of the particle at time t .

(2)

- (ii) State the range of values of the acceleration of the particle.

(3)

- (b) When $t = 0$, the particle is at the origin.

Find an expression for the displacement of the particle from the origin at time t .

(4)

(Total 9 marks)

Q29.

A particle moves such that at time t seconds its acceleration is given by

$$(2 \cos t \mathbf{i} - 5 \sin t \mathbf{j}) \text{ m s}^{-2}$$

- (a) The mass of the particle is 6 kg. Find the magnitude of the resultant force on the particle when $t = 0$.

(3)

- (b) When $t = 0$, the velocity of the particle is $(2\mathbf{i} + 10\mathbf{j}) \text{ m s}^{-1}$.

Find an expression for the velocity of the particle at time t .

(5)

(Total 8 marks)

Q30.

A particle moves in a horizontal plane, in which the unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively. At time t seconds, its position vector, $\mathbf{r}$ metres, is given by

$$\mathbf{r} = (2t^3 - t^2 + 6)\mathbf{i} + (8 - 4t^3 + t)\mathbf{j}$$

- (a) Find an expression for the velocity of the particle at time t .

(3)

- (b) (i) Find the velocity of the particle when $t = \frac{1}{3}$.

(2)

(ii) State the direction in which the particle is travelling at this time.

(1)

(c) Find the acceleration of the particle when $t = 4$.

(3)

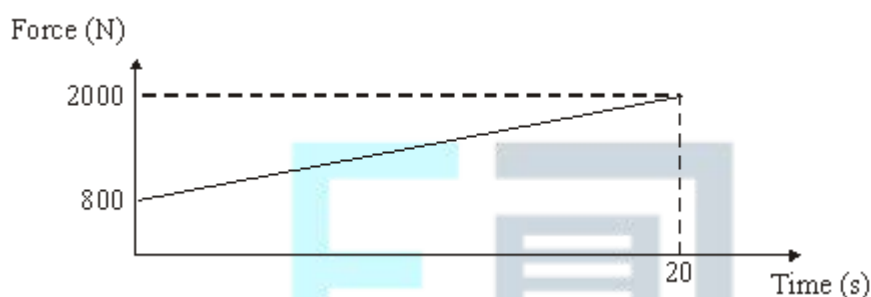
(d) The mass of the particle is 6 kg. Find the magnitude of the resultant force on the particle when $t = 4$.

(3)

(Total 12 marks)

Q31.

The graph shows a model for the resultant horizontal force on a car, which varies as it accelerates from rest for 20 seconds. The mass of the car is 1200 kg.



(a) The acceleration of the car at time t seconds is $a \text{ m s}^{-2}$. Show that

$$a = \frac{2}{3} + \frac{t}{20}, \text{ for } 0 \leq t \leq 20$$

(5)

(b) Find an expression for the velocity of the car at time t .

(3)

(c) Find the distance travelled by the car in the 20 seconds.

(4)

(d) An alternative model assumes that the resultant force increases uniformly from 900 to 2100 newtons during the 20 seconds. Which term in your expression for the velocity would change as a result of this modification? Explain why.

(2)

(Total 14 marks)

Q32.

A particle P , of mass 0.25 kg, moves in a horizontal plane so that at time t seconds its velocity $\mathbf{v} \text{ m s}^{-1}$ is given by

$$\mathbf{v} = 8 \sin t \mathbf{i} + 4 \cos t \mathbf{j}$$

- (a) Find the force, $\mathbf{F}$ newtons, acting on P at time t . (3)

- (b) (i) Show that the magnitude of $\mathbf{F}$ at time t is

$$\sqrt{(3 \cos^2 t + 1)}$$

(3)

- (ii) Hence state the range of values of the magnitude of $\mathbf{F}$.

(2)

(Total 8 marks)

Q33.

A particle P moves in a horizontal plane so that at time t seconds after the motion begins its position vector $\mathbf{r}$ metres is given by

$$\mathbf{r} = \sin 2t \mathbf{i} + 6t \mathbf{j}$$

- (a) (i) Find the velocity of P at time t . (2)

- (ii) Show that the speed of P at time t is

$$\sqrt{(4 \cos^2 2t + 36)}$$

(3)

- (iii) Find the first value of t for which the speed of P has its minimum value.

(2)

- (b) (i) The mass of P is 0.25 kg. Find the force, $\mathbf{F}$ newtons, acting on P at time t . (3)

- (ii) State the direction of $\mathbf{F}$ and explain why its magnitude never exceeds one newton.

(2)

(Total 12 marks)

Q34.

A particle P , of mass 0.5 kg, moves under the action of a force $\mathbf{F}$ newtons, where $\mathbf{F} = 2t\mathbf{i} + 1.5\mathbf{j}$ at time t seconds.

- (a) Find the acceleration of P at time t . (1)

- (b) The particle P starts its motion from rest.

Find the velocity of P at time t .

(2)

(c) The particle P starts its motion at the point O .

(i) Find the position vector of P relative to O at time t .

(3)

(ii) Find the distance of P from O when it is moving parallel to the vector $\mathbf{i} + \mathbf{j}$.

(5)

(Total 11 marks)

Q35.

A particle P moves so that at time t seconds its position vector, $\mathbf{r}$ metres, is given by

$$\mathbf{r} = 6t\mathbf{i} + t^2\mathbf{j}.$$

(a) Find the velocity of P at time t .

(2)

(b) Find an expression for the speed of P at time t .

(2)

(c) Find the value of t when P moves with speed $6\sqrt{2}$ m s⁻¹.

(2)

(Total 6 marks)